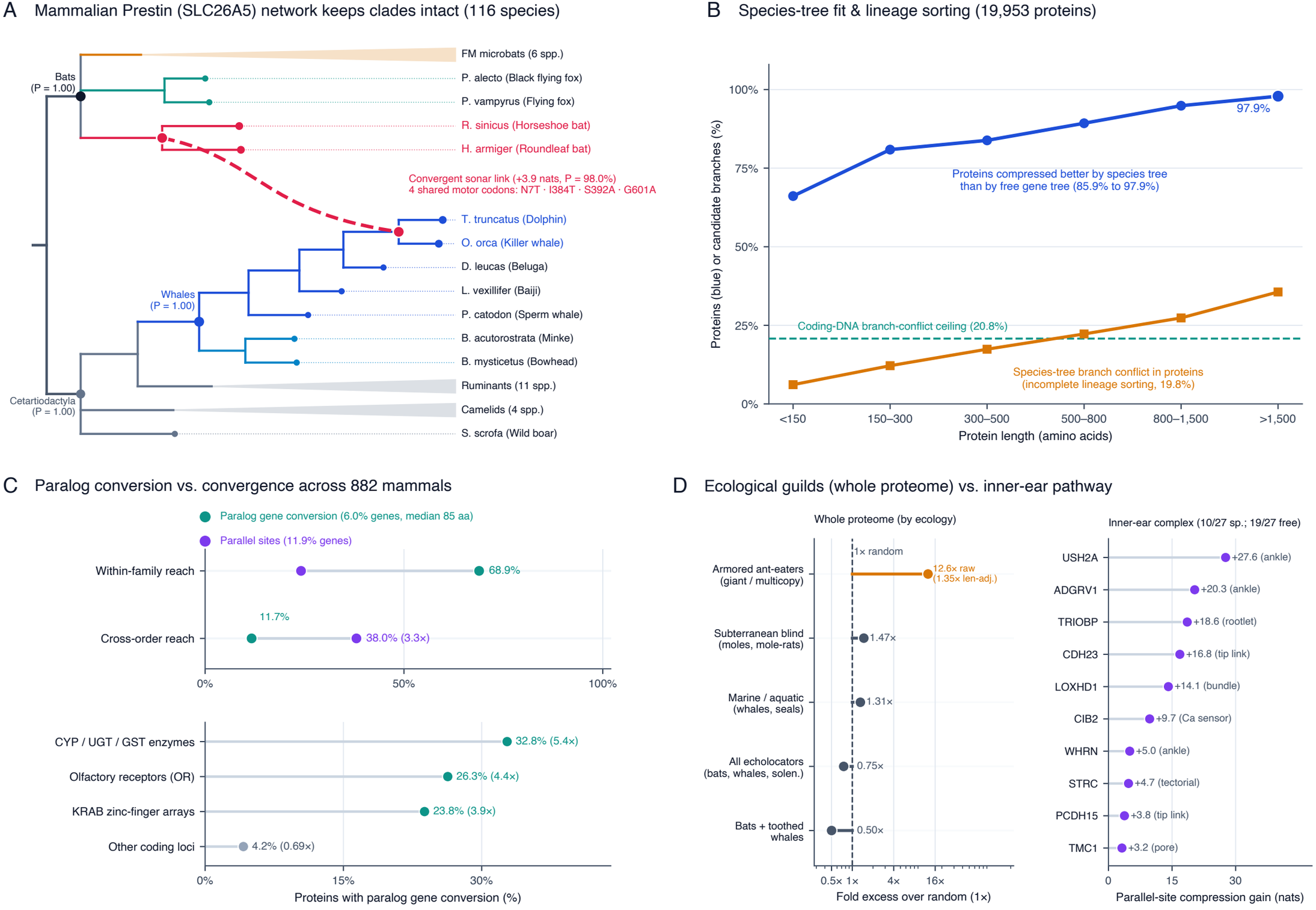
A
Mammalian Prestin (SLC26A5) network keeps clades intact (116 species)
Bats
(P = 1.00)
FM microbats (6 spp.)
P. alecto (Black flying fox)
P. vampyrus (Flying fox)
R. sinicus (Horseshoe bat)
H. armiger (Roundleaf bat)
Convergent sonar link (+3.9 nats, P = 98.0%)
4 shared motor codons: N7T · I384T · S392A · G601A
T. truncatus (Dolphin)
O. orca (Killer whale)
D. leucas (Beluga)
L. vexillifer (Baiji)
Whales
(P = 1.00)
P. catodon (Sperm whale)
B. acutorostrata (Minke)
B. mysticetus (Bowhead)
Ruminants (11 spp.)
Cetartiodactyla
(P = 1.00)
Camelids (4 spp.)
S. scrofa (Wild boar)
B
Species-tree fit & lineage sorting (19,953 proteins)
Proteins (blue) or candidate branches (%)
100%
75%
50%
25%
0%
97.9%
Proteins compressed better by species tree
than by free gene tree (85.9% to 97.9%)
Coding-DNA branch-conflict ceiling (20.8%)
Species-tree branch conflict in proteins
(incomplete lineage sorting, 19.8%)
<150
150–300
300–500
500–800
800–1,500
>1,500
Protein length (amino acids)
C
Paralog conversion vs. convergence across 882 mammals
Paralog gene conversion (6.0% genes, median 85 aa)
Parallel sites (11.9% genes)
Within-family reach
68.9%
11.7%
Cross-order reach
38.0% (3.3×)
0%
50%
100%
CYP / UGT / GST enzymes
32.8% (5.4×)
Olfactory receptors (OR)
26.3% (4.4×)
KRAB zinc-finger arrays
23.8% (3.9×)
Other coding loci
4.2% (0.69×)
0%
15%
30%
Proteins with paralog gene conversion (%)
D
Ecological guilds (whole proteome) vs. inner-ear pathway
Whole proteome (by ecology)
1× random
Armored ant-eaters
(giant / multicopy)
12.6× raw
(1.35× len-adj.)
Subterranean blind
(moles, mole-rats)
1.47×
Marine / aquatic
(whales, seals)
1.31×
All echolocators
(bats, whales, solen.)
0.75×
Bats + toothed
whales
0.50×
0.5×
1×
4×
16×
Fold excess over random (1×)
Inner-ear complex (10/27 sp.; 19/27 free)
USH2A
+27.6 (ankle)
ADGRV1
+20.3 (ankle)
TRIOBP
+18.6 (rootlet)
CDH23
+16.8 (tip link)
LOXHD1
+14.1 (bundle)
CIB2
+9.7 (Ca sensor)
WHRN
+5.0 (ankle)
STRC
+4.7 (tectorial)
PCDH15
+3.8 (tip link)
TMC1
+3.2 (pore)
0
15
30
Parallel-site compression gain (nats)